DRESS/DiHS syndrome induced by Propylthiouracil: a case report.

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Drug reaction with eosinophilia and systemic symptoms (DRESS), also known as Drug-induced hypersensitivity syndrome (DiHS), is a severe adverse drug reaction. Propylthiouracil, a member of thiouracils group, is widely used in medical treatment of hyperthyroidism. Propylthiouracil is associated with multiple adverse effects such as rash, agranulocytosis hepatitis and antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis, but rarely triggers DRESS/DiHS syndrome. Here, we describe a severe case of propylthiouracil-induced DRESS/DiHS syndrome. A 38-year-old female was treated with methimazole for hyperthyroidism at first. 4 weeks later, the patient developed elevated liver transaminase so methimazole was stopped. After liver function improved in 2 weeks, medication was switched to propylthiouracil therapy. The patient subsequently developed nausea and rash followed by a high fever, acute toxic hepatitis and multiple organ dysfunction (liver, lung and heart), which lasted for 1 month after propylthiouracil was started. According to the diagnostic criteria, the patient was diagnosed of DRESS/DiHS syndrome which was induced by propylthiouracil. As a result, propylthiouracil was immediately withdrawn. And patient was then treated with adalimumab, systematic corticosteroids and plasmapheresis in sequence. Symptoms were finally resolved 4 weeks later. Propylthiouracil is a rare cause of the DRESS/DiHS syndrome, which typically consists of severe dermatitis and various degrees of internal organ involvement. We want to emphasize through this severe case that DRESS/DiHS syndrome should be promptly recognized to hasten recovery.

Prolonged Drug-Induced Hypersensitivity Syndrome/DRESS With Alopecia Areata and Autoimmune Thyroiditis.

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Drug-induced hypersensitivity syndrome (DIHS), also called drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome, is a potentially fatal drug-induced hypersensitivity reaction that is characterized by a cutaneous eruption, multiorgan involvement, viral reactivation, and hematologic abnormalities. We present a case of lamotrigine-associated DIHS/DRESS complicated by an unusually prolonged course requiring oral corticosteroids and narrow-band ultraviolet B treatment and with development of extensive alopecia areata and autoimmune thyroiditis. DIHS/DRESS is a severe cutaneous adverse reaction that may require prolonged treatment until symptoms resolve. Oral corticosteroids are the mainstay of treatment, but long-term use is associated with significant adverse effects. Alternative therapies, such as cyclosporine, look promising, but further studies are needed to determine safety profile and efficacy. DIHS/DRESS patients also should be educated and followed for potential autoimmune sequelae.

Drug Reaction With Eosinophilia and Systemic Symptoms (DRESS) Syndrome to Hemodialysis Polysulfone Membrane.

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Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome is a severe and potentially life-threatening hypersensitivity reaction. Although commonly associated with specific drugs, there have been no reports of DRESS syndrome caused by medical devices. We report a unique case of DRESS syndrome linked to a particular hemodialysis membrane during treatment. An 83-year-old man on hemodialysis exhibited fever, rash, and elevated eosinophils. Despite medication changes and consultations with specialists, his condition persisted. A drug-induced lymphocyte stimulation test revealed a positive response to the dialysis membrane. His symptoms and lab results met DRESS syndrome diagnostic criteria. After substituting the membrane and administering glucocorticoids, the patient displayed early improvement. Diagnosing DRESS syndrome is complex due to its varied presentation and lack of specific benchmarks. This instance underscores the need to consider medical devices as potential DRESS syndrome triggers. Enhanced physician awareness can facilitate prompt detection and proper management, ultimately refining patient outcomes.

Allopurinol-Induced Drug Reaction With Eosinophilia and Systemic Symptoms: A Case Report.

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Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome is an uncommon yet serious adverse cutaneous drug reaction that results from a hypersensitivity reaction. Drug reaction with eosinophilia and systemic symptoms is often misdiagnosed because of vague and confounding signs and symptoms. The most common clinical manifestations of DRESS are shared with many other diseases and include rash, lymphadenopathy, and fever. Because the syndrome can be difficult to diagnose, patients are often in the late stages of the disease process before treatment is initiated. The mainstay of treatment is stopping the culprit medication. Drug reaction with eosinophilia and systemic symptoms is associated with a high mortality rate, most often from liver failure and failure to diagnose. Emergency providers should be able to recognize the clinical manifestations of DRESS, know what diagnostic studies are indicated, and be familiar with the appropriate treatment.

Drug Reaction with Eosinophilia and Systemic Symptoms Syndrome Induced by Levetiracetam in a Pediatric Patient.

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Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome is a rare, life-threatening hypersensitivity drug reaction. Patients present with cutaneous rash, fever, lymphadenopathy, hematologic abnormalities with eosinophilia and atypical lymphocytes, and visceral organ involvement. The prognosis of DRESS syndrome is related to the degree of end-organ damage, and the mortality rate is approximately 10%. We report a 9-year-old girl treated with only levetiracetam because of intracranial space occupying mass-related seizures. The patient developed pharyngitis accompanied by exudative membrane, bilateral cervical lymphadenopathy, tender hepatomegaly, skin rash, and fever after 19 days of levetiracetam therapy. Laboratory findings revealed leukocytosis, lymphocytosis with an atypical lymphocytosis, eosinophilia, thrombocytopenia, and elevated serum transaminases. Serologic studies of viruses were negative. The patient was diagnosed with DRESS syndrome and antiepileptic therapy was ceased immediately. The systemic signs and symptoms of the patient were improved after systemic steroid and antihistamine therapy. WHY SHOULD AN EMERGENCY PHYSICIAN BE AWARE OF THIS?: It is important that emergency physicians be aware of the possibility of DRESS syndrome when attending children that present with clinical viral infections. We would like to emphasize that obtaining a careful and detailed medication history is an essential part of clinical assessment for the diagnosis of DRESS syndrome.

Results of a randomized double-blind study evaluating luvadaxistat in adults with Friedreich ataxia.

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Friedreich ataxia (FRDA) is a rare disorder with progressive neurodegeneration and cardiomyopathy. Luvadaxistat (also known as TAK-831; NBI-1065844), an inhibitor of the enzyme d-amino acid oxidase, has demonstrated beneficial effects in preclinical models relevant to FRDA. This phase 2, randomized, double-blind, placebo-controlled, parallel-arm study evaluated the efficacy and safety of oral luvadaxistat in adults with FRDA. Adult patients with FRDA were randomized 2:1:2 to placebo, luvadaxistat 75 mg twice daily (BID), or luvadaxistat 300 mg BID for 12 weeks. The primary endpoint changed from baseline at week 12 on the inverse of the time to complete the nine-hole peg test (9-HPT-1), a performance-based measure of the function of the upper extremities and manual dexterity. Comparisons between luvadaxistat and placebo were made using a mixed model for repeated measures. Of 67 randomized patients, 63 (94%) completed the study. For the primary endpoint, there was no statistically significant difference in change from baseline on the 9-HPT-1 (seconds-1) at week 12 between placebo (0.00029) and luvadaxistat 75 mg BID (-0.00031) or luvadaxistat 300 mg BID (-0.00059); least squares mean differences versus placebo (standard error) were -0.00054 (0.000746) for the 75 mg dose and -0.00069 (0.000616) for the 300 mg dose. Luvadaxistat was safe and well tolerated; the majority of reported adverse events were mild in intensity. Luvadaxistat was safe and well tolerated in this cohort of adults with FRDA; however, it did not demonstrate efficacy as a treatment for this condition.

EGFR/EGFRvIII-targeted immunotoxin therapy for the treatment of glioblastomas via convection-enhanced delivery.

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Glioblastoma is the most aggressive malignant brain tumor among all primary brain and central nervous system tumors. The median survival time for glioblastoma patients given the current standard of care treatment (surgery, radiation, and chemotherapy) is less than 15 months. Thus, there is an urgent need to develop more efficient therapeutics to improve the poor survival rates of patients with glioblastoma. To address this need, we have developed a novel tumor-targeted immunotoxin (IT), D2C7-(scdsFv)-PE38KDEL (D2C7-IT), by fusing the single chain variable fragment (scFv) from the D2C7 monoclonal antibody (mAb) with the Pseudomonas Exotoxin (PE38KDEL). D2C7-IT reacts with both the wild-type epidermal growth factor receptor (EGFRwt) and EGFR variant III (EGFRvIII), two onco-proteins frequently amplified or overexpressed in glioblastomas. Surface plasmon resonance and flow cytometry analyses demonstrated a significant binding capacity of D2C7-IT to both EGFRwt and EGFRvIII proteins. In vitro cytotoxicity data showed that D2C7-IT can effectively inhibit protein synthesis and kill a variety of EGFRwt-, EGFRvIII-, and both EGFRwt- and EGFRvIII-expressing glioblastoma xenograft cells and human tumor cell lines. Furthermore, D2C7-IT exhibited a robust anti-tumor efficacy in orthotopic mouse glioma models when administered via intracerebral convection-enhanced delivery (CED). A preclinical toxicity study was therefore conducted to determine the maximum tolerated dose (MTD) and no-observed-adverse-effect-level (NOAEL) of D2C7-IT via intracerebral CED for 72 hours in rats. Based on this successful rat toxicity study, an Investigational New Drug (IND) application (#116855) was approved by the Food and Drug Administration (FDA), and is now in effect for a Phase I/II D2C7-IT clinical trial (D2C7 for Adult Patients with Recurrent Malignant Glioma, https://clinicaltrials.gov/ct2/show/NCT02303678). While it is still too early to draw conclusions from the trial, results thus far are promising.

Towards personalized therapy for patients with glioblastoma.

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Combined therapy with temozolomide and radiotherapy is a standard treatment and improves the survival for patients with newly diagnosed glioblastoma. However, the prognosis remains poor, with a median survival time of 12-15 months. Currently, several clinical trials of dose-dense temozolomide regimen or molecular-targeting therapies have been performed to overcome the resistance of glioblastoma. In these therapies, rational prognostic biomarkers have also been investigated to predict their outcome and response to treatment. This advanced understanding of the biological markers can help to develop personalized therapies for glioblastoma patients. Generally, due to a reduced tolerance, elderly patients do not seem to benefit from intensive treatment. This population needs individual treatments depended on their age or performance status. In this article, we review the recent studies that can provide personalized therapy for glioblastoma, based on molecular tumor profiling or patients' physical status.

Key Clinical Principles in the Management of Glioblastoma.

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Glioblastoma is the most common and aggressive primary brain tumor in the adult population and leads to considerable morbidity and mortality. It has a dismal prognosis with average survival of 15-18 months, and the current standard-of-care treatment paradigm includes maximal surgical resection and postoperative concurrent chemoradiotherapy and maintenance chemotherapy, with consideration of Tumor Treating Fields. There is a major emphasis to enroll patients onto ongoing clinical trials to further improve treatment outcomes, given the aggressive nature of the disease course and poor patient survival. Recent research efforts have focused on radiotherapy dose intensification, regulation of the tumor microenvironment, and exploration of immunotherapeutic approaches to overcome the barriers to treatment. This review article outlines the current evidence-based management principles as well as reviews recent clinical trial data and ongoing clinical studies evaluating novel therapeutic options.

Delaying standard combined chemoradiotherapy after surgical resection does not impact survival in newly diagnosed glioblastoma patients.

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To assess the influence of the time interval between surgical resection and standard combined chemoradiotherapy on survival in newly diagnosed and homogeneously treated (surgical resection plus standard combined chemoradiotherapy) glioblastoma patients; while controlling confounding factors (extent of resection, carmustine wafer implantation, functional status, neurological deficit, and postoperative complications). From 2005 to 2011, 692 adult patients (434 men; mean of 57.5 ± 10.8 years) with a newly diagnosed glioblastoma were enrolled in this retrospective multicentric study. All patients were treated by surgical resection (65.5% total/subtotal resection, 34.5% partial resection; 36.7% carmustine wafer implantation) followed by standard combined chemoradiotherapy (radiotherapy at a median dose of 60 Gy, with daily concomitant and adjuvant temozolomide). Time interval to standard combined chemoradiotherapy was analyzed as a continuous variable and as a dichotomized variable using median and quartiles thresholds. Multivariate analyses using Cox modeling were conducted. The median progression-free survival was 10.3 months (95% CI, 10.0-11.0). The median overall survival was 19.7 months (95% CI, 18.5-21.0). The median time to initiation of combined chemoradiotherapy was 1.5 months (25% quartile, 1.0; 75% quartile, 2.2; range, 0.1-9.0). On univariate and multivariate analyses, OS and PFS were not significantly influenced by time intervals to adjuvant treatments. On multivariate analysis, female gender, total/subtotal resection and RTOG-RPA classes 3 and 4 were significant independent predictors of improved OS. Delaying standard combined chemoradiotherapy following surgical resection of newly diagnosed glioblastoma in adult patients does not impact survival.